

Case Study 1: Morphological Typology of Baltic and Tocharian

Morphological systems are often described as a spectrum with fusional morphology on one end and agglutinative morphology on the other (Haspelmath & Sims, 2010). Fusional systems work by encoding multiple pieces of information (i.e. case, number, gender, etc.) in a single morpheme. In other words, those meanings are fused together into one form.

Agglutinative systems, on the other hand, work by giving each meaning its own distinct morpheme. Proto-Indo-European (PIE) in its reconstructed form is thought to be highly fusional, and many of its branches exhibit this characteristic (Klein et al., 2017). In this paper, the morphological typology of the Tocharian and Baltic branches of PIE will be examined. These branches are particularly interesting since Tocharian has notably diverged from PIE and demonstrates agglutinative capabilities. Baltic (chiefly Lithuanian), on the other hand, is considered a conservative branch in this regard and likely reflects PIE fusionalty. The development of agglutinative morphology in Tocharian is theorized to have emerged from Uralic or Altaic language contact, although it is still feasible that it occurred internally.

First, this section explores how the Lithuanian morphological system works. Lithuanian has two genders (masculine and feminine), two numbers (singular and plural), and seven noun cases (nominative, genitive, dative, accusative, instrumental, locative, and vocative) (Ambrasas et al., 1997). While Lithuanian is missing the neuter gender and dual number from PIE, it has maintained seven of the eight original PIE cases (Klein et al., 2017). The only missing case is ablative. The following examples show how case, number, and gender are marked in Lithuanian:

Lithuanian

- | | | |
|-----|---------------------|-----------------|
| (1) | ger-as | tėv-as |
| | good-NOM.SG.M | father-NOM.SG.M |
| | ‘a good father’ | |
| (2) | ger-a | motin-a |
| | good-NOM.SG.F | mother-NOM.SG.F |
| | ‘a good mother’ | |
| (3) | ger-ais | tėv-ais |
| | good-INS.PL.M | father-INS.PL.M |
| | ‘with good fathers’ | |
| (4) | ger-omis | motin-omis |
| | good-INS.PL.F | mother-INS.PL.F |
| | ‘with good mothers’ | |

(Ambrazas et al., 1997)

All of the examples above demonstrate fusional morphology. Case, number, and gender are conveyed by the same morpheme. If you compare examples 1 and 3, for instance, it is not possible to further divide the final morpheme into a form signifying instructive and a form signifying plural. The single form fuses all of the meanings together. Remarkably the morpheme is the same whether it is applied to a noun or an adjective, and they both agree.

Although Lithuanian is the prime example of preserved fusional morphology from PIE, it is worth noting that Old Lithuanian, spoken from the 16th to late 19th century, also establishes probable Indo-European precedent for acquiring agglutination through language contact, even from Finno-Ugric languages (Fortson, 2010). In the Bronze Age, the Balts had extended contact with Finnic tribes, as is evident from Baltic borrowings to Finnic languages as well. The following examples illustrate Finnic-like agglutinative constructions for illative and allative cases in Old Lithuanian:

Old Lithuanian

- (5) nam-uos-na tavu-uos-na
house-M.ACC.PL-ILL your-M.ACC.PL-ILL
'(up to) into your house'
- (6) pagirt-os-p merg-os-p Mari-os-p
praised-F.GEN.SG-ALL maiden-F.GEN.SG-ALL Maria-F.GEN.SG-ALL
'to the praised virgin Maria'

(Beekes, 2011)

The illative, allative, adessive, and inessive (cases present in Finnic languages) were created by starting with a standard fusional ending like *-uos* (accusative) and appending a grammaticalized postposition like *-na* (illative) (Beekes, 2011). These forms eventually disappeared in modern Lithuanian except in frozen terms like *velnióp* ('to hell'), but they set a precedent nevertheless (Ambrazas et al., 1997).

This section will explore how the Tocharian branch's morphological system works. Tocharian is attested in writings from the first millennium CE, but is now extinct (Klein et al., 2017). Tocharian retained all three numbers from PIE (singular, dual, and plural) and innovated two new ones as well, par and plurative (Fortson, 2010). Tocharian also retained the three PIE genders (masculine, feminine, and neuter), but it lost five of the PIE cases leaving just nominative, vocative (only in Tocharian B), genitive, and accusative (called oblique). These are known as the primary cases. Then, Tocharian innovated agglutinative cases in a very similar way to Old Lithuanian, which suggests that it could have also come

about as a result of language contact. These agglutinative cases are known as secondary cases, since they are always paired with a primary case first. Perlative, comitative, allative, ablative, and locative cases are present in both Tocharian A and B. The instrumental case is present as a secondary case only in Tocharian A, and casual only in Tocharian B. In other words, Tocharian was very productive in creating these agglutinative cases. The following examples show some of Tocharian's agglutinative secondary cases in use:

Tocharian B

(7) pā-tär

father-M.OBL.SG

‘the father’

(8) pā-tär-mpa

father-M.OBL.SG-COM

‘with the father’

(9) pā-tär-mem

father-M.OBL.SG-ABL

‘from the father’

(Fortson, 2010)

Just like in Old Lithuanian, these agglutinative cases appear after a standard fusional ending. The secondary cases convey only the meaning of case and do not react to the number of the noun it is attached to (Klein et al., 2017). These morphemes also developed from previously separate postpositions.

The Indo-Aryan branch of Indo-European was able to develop an agglutinative case system based on postpositions gradually without any contact-related influence (Klein et al., 2017), so how can it be known for sure that Tocharian did not develop in the same way? Firstly, the Tocharian system is much more extensive than the Indo-Aryan one. While Tocharian's SOV word order, like in Indo-Aryan, was well-suited for grammaticalizing postpositions, it likely would not have produced six or seven agglutinative cases on its own as had happened. Tocharian syntax was simply open to influence in that way. In other words, internal factors probably contributed to the change, but were not sufficient to produce such a large effect.

It is also notable that the certain cases which were innovated in Tocharian were Uralic in nature, such as the perlative and comitative (Peyrot, 2019). The perlative in particular is found in many Uralic languages. The lack of a dative case also matches Uralic paradigms. These phenomena especially point to some kind of Uralic influence in particular, although all

candidate contact languages in the Tocharian region except Old Iranian exhibit an agglutinative case system. So, there were many opportunities for contact, which increases the overall likelihood.

Borrowability hierarchies could also support the hypothesis that Tocharian's agglutinative secondary case system was influenced by language contact, with a caveat. Thomason and Kaufman's well-regarded borrowability hierarchy regards inflectional morphology as very resistant and thus requires fairly intense contact to change (as cited in Gardani, 2021). However, lexical borrowing, which happens at casual levels of contact, did not happen between Uralic and Tocharian (Peyrot, 2019), unlike with Uralic and Old Lithuanian. Nevertheless, if a result of substrate influence, the entire hierarchy is flipped, and inflectional influence without lexical borrowing becomes much more likely. Substrate influence occurs when, for instance, a lower-prestige population of Uralic language speakers attempt to master higher-prestige Tocharian and end up imposing lower-prestige language qualities, like inflectional patterns, onto the higher-prestige language. This scenario could fit the Tocharian situation nicely.

The languages of the Indo-European family exhibit lots of variation. Some, like Lithuanian, have preserved PIE features like its case system very well. Others, like Tocharian A and B, ended up with a drastically different case system. A fairly compelling case has been proposed for Uralic or other language contact leading Tocharian to undergo massive change in its inflectional system. There is evidence in other branches of Indo-European that prove this is possible. However, many conditions (like the presence of a Uralic substrate) need to be in place for the theory to hold. Perhaps these conditions are too demanding to be sustained and another explanation is needed.